# Novelty Analysis

Analysis of Novelty, Transience, and Resonance

PENPAL Project

2026-02-22

## Contents


0.1 Summary . . . 1

**1 1. Data Loading and Preprocessing 1**

1.1 1.1 Load Data . . . 1

1.2 1.2 Reshape Data . . . 2

1.3 1.3 Descriptive Statistics . . . 2

**2 2. Novelty and Resonance Analysis 4**

2.1 2.1 Resonance vs Novelty Plot . . . 4

2.2 2.2 Transience vs Novelty Plot . . . 5

**3 3. Statistical Models 6**

3.1 3.1 Resonance Model . . . 6

3.2 3.2 Transience Model . . . 7

**4 4. Session Information 7**


### 0.1 Summary

- Humans produce higher novelty (surprisal) and higher transience than the model, indicating stronger deviations from earlier story context.
- Both agents display a positive novelty–resonance relationship (innovation bias), but the slope is steeper for humans.
- This indicates that surprising human turns exert greater forward influence on narrative development.
- AI contributions show lower variance and lower informational impact relative to human inputs.
- Humans are the primary drivers of narrative innovation and thematic redirection.

# 1 1. Data Loading and Preprocessing

## 1.1 1.1 Load Data

```
# Load Field Data
df_res_field <- read_csv(here("data/TEXT/processed/novelty_scores_40.csv")) %>%
  filter(!is.na(user_surprise) & !is.na(ai_surprise)) %>%
  mutate(
    user_resonance = user_surprise - user_transience,
    ai_resonance = ai_surprise - ai_transience,
    datatype = "Field"
```

```r
  ) %>%
  drop_na(user, ai) %>%
  filter(interaction_count > 1, interaction_count < 10, ai_surprise < 0)
```

## 1.2 1.2 Reshape Data

```r
df_res_long <- df_res_field %>%
  pivot_longer(
    cols = c(user_surprise, user_transience, ai_surprise, ai_transience),
    names_to = c("agent", "metric"),
    names_pattern = "(user|ai)_(surprise|transience)"
  ) %>%

  pivot_wider(names_from = metric, values_from = value) %>%
  mutate(
    resonance = surprise - transience,
    agent = factor(agent, levels = c("user", "ai"), labels = c("User", "LLM"))
  )
```

## 1.3 1.3 Descriptive Statistics

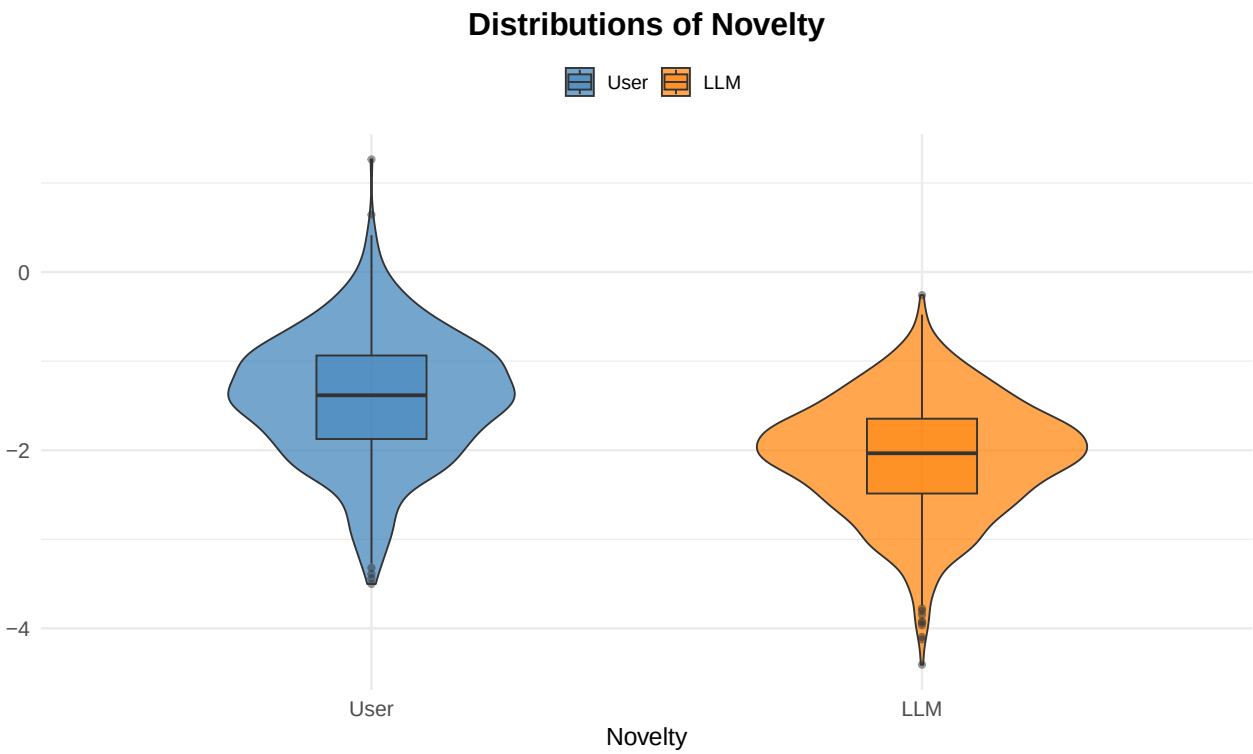
```r
desc_stats <- df_res_long %>%
  group_by(agent) %>%
  summarise(
    n = n(),
    mean_surprise = mean(surprise, na.rm = TRUE),
    sd_surprise = sd(surprise, na.rm = TRUE),
    mean_transience = mean(transience, na.rm = TRUE),
    sd_transience = sd(transience, na.rm = TRUE),
    mean_resonance = mean(resonance, na.rm = TRUE),
    sd_resonance = sd(resonance, na.rm = TRUE)
  )

knitr::kable(desc_stats, caption = "Descriptive Statistics for Novelty Metrics by Agent", digits = 3)
```

Table 1: Descriptive Statistics for Novelty Metrics by Agent

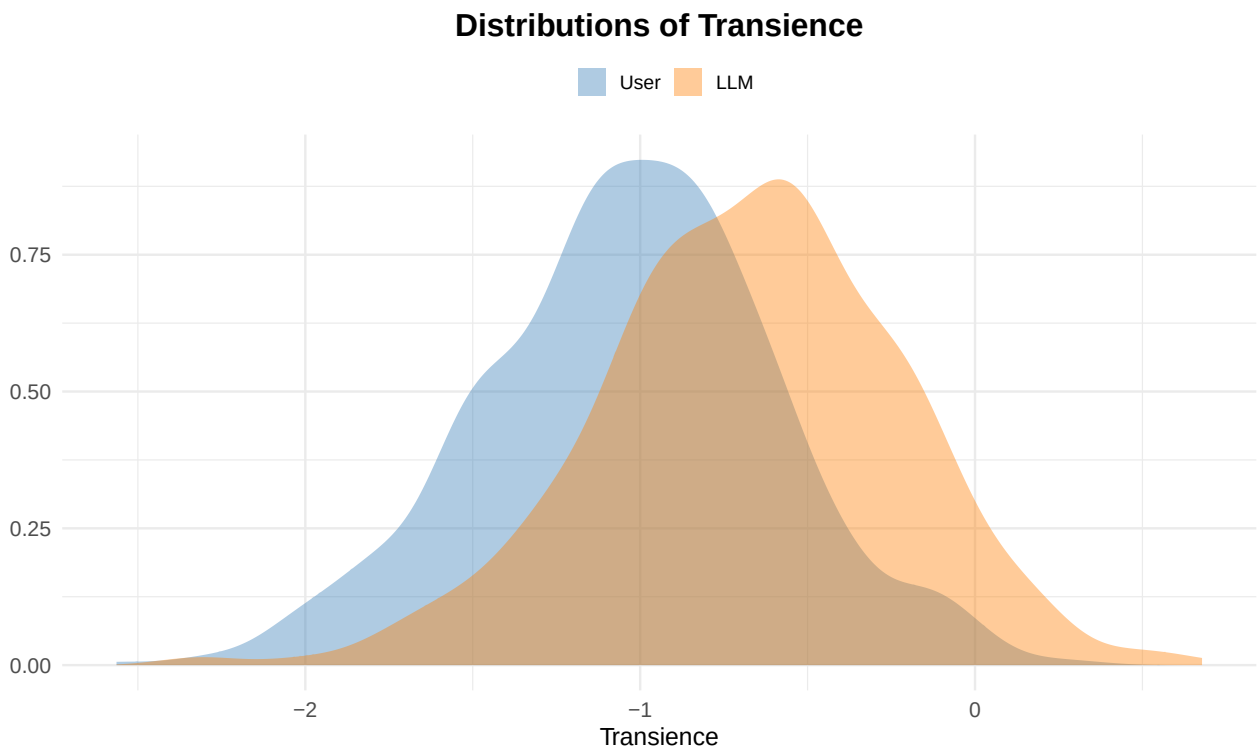
| agent | n | mean_surprise | sd_surprise | mean_transience | sd_transience | mean_resonance | sd_resonance |
|---|---|---|---|---|---|---|---|
| User | 700 | -1.418 | 0.713 | -1.034 | 0.441 | -0.384 | 0.774 |
| LLM | 700 | -2.077 | 0.647 | -0.676 | 0.454 | -1.401 | 0.684 |

```r
p_novelty_agent <- ggplot(df_res_long, aes(x = agent, y = surprise, fill = agent)) +
  geom_violin(width = 0.6, alpha = 0.7) +
  geom_boxplot(width = 0.2, alpha = 0.5) +
  scale_fill_manual(values = c("User" = "#377EB8", "LLM" = "#FF7F00"), name = "") +
  scale_color_manual(values = c("User" = "#377EB8", "LLM" = "#FF7F00")) +
  labs(title = "Distributions of Novelty", x = "Novelty", y = NULL) +
  theme(
    legend.position = "top",
    plot.title = element_text(hjust = 0.5, size = 18, face = "bold"),
    axis.text = element_text(size = 12)
```

```
  )
print(p_novelty_agent)
```




```
ggsave(
  file.path(GRAPHS_DIR, "Surprise_Distributions_Agent.png"),
  p_novelty_agent,
  width = 10,
  height = 6,
  dpi = 300
)

p_transience_agent <- ggplot(df_res_long, aes(x = transience, fill = agent)) +
  geom_density(alpha = 0.4, color = NA) +
  scale_fill_manual(values = c("User" = "#377EB8", "LLM" = "#FF7F00"), name = "") +
  scale_color_manual(values = c("User" = "#377EB8", "LLM" = "#FF7F00")) +
  labs(title = "Distributions of Transience", x = "Transience", y = NULL) +
  theme(
    legend.position = "top",
    plot.title = element_text(hjust = 0.5, size = 18, face = "bold"),
    axis.text = element_text(size = 12)
  )
print(p_transience_agent)
```

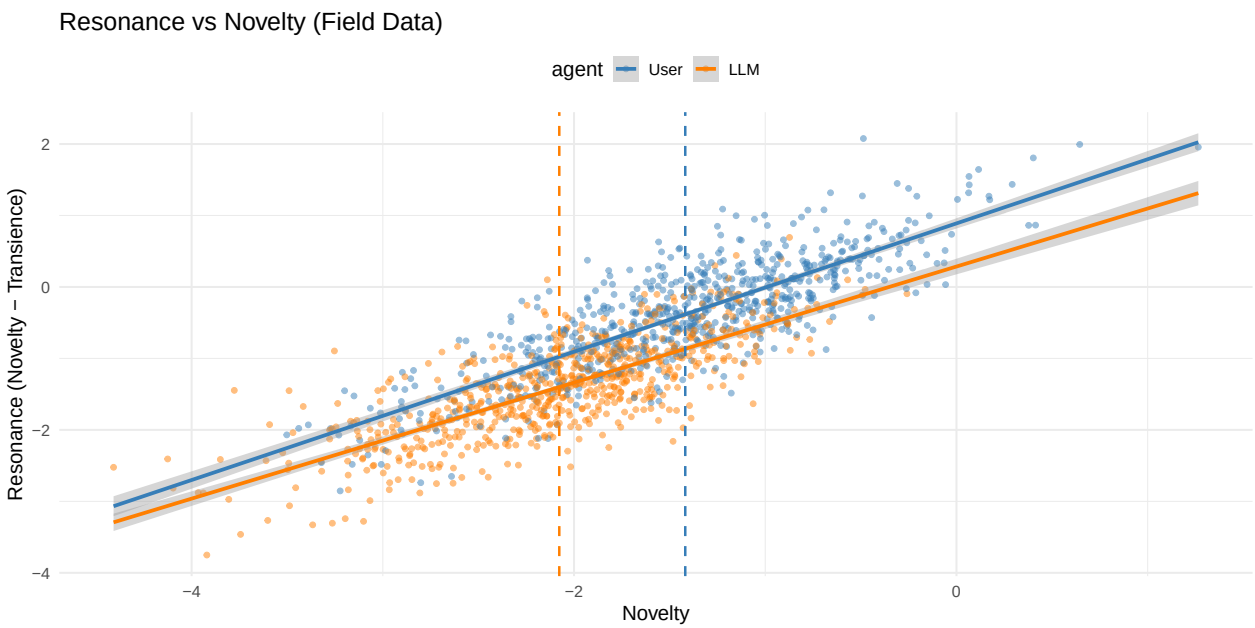



```
ggsave(
  file.path(GRAPHS_DIR, "Transience_Distributions_Agent.png"),
  p_transience_agent,
  width = 10,
  height = 6,
  dpi = 300
)
```

# 2 2. Novelty and Resonance Analysis

## 2.1 2.1 Resonance vs Novelty Plot

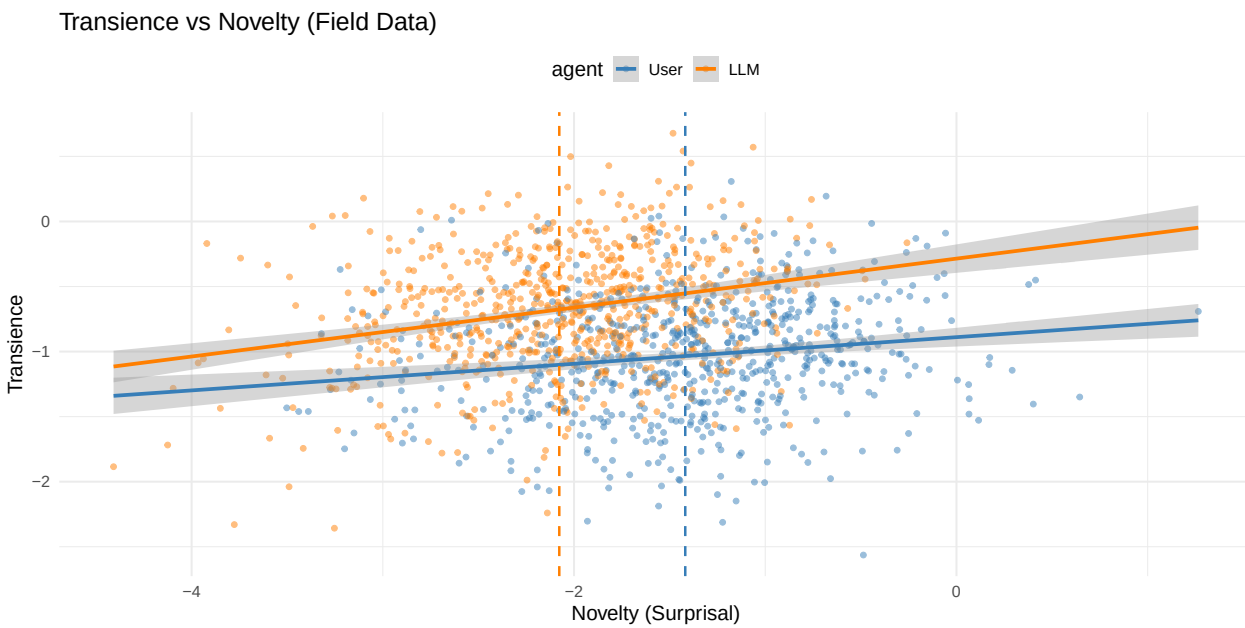
```
# Plot Resonance vs Novelty (Field Data)
p_resonance <- ggplot(df_res_long, aes(x = surprise, y = resonance, color = agent)) +
  geom_point(alpha = 0.5, size = 1.5) +
  geom_smooth(method = "lm", se = TRUE, fullrange = TRUE, size = 1.2) +
  scale_color_manual(values = c("User" = "#377EB8", "LLM" = "#FF7F00")) +
    geom_vline(data = desc_stats,
               aes(xintercept = mean_surprise, color = agent),
               linetype = "dashed", linewidth = 0.8, show.legend = FALSE) +
  labs(
    title = "Resonance vs Novelty (Field Data)",
    x = "Novelty",
    y = "Resonance (Novelty - Transience)"
  ) +
  theme_minimal(base_size = 14) +
  theme(legend.position = "top")

print(p_resonance)
```




```
ggsave(file.path(GRAPHS_DIR, "novelty_resonance_vs_novelty.png"),
       p_resonance, width = 12, height = 6, dpi = 300)
```

## 2.2 2.2 Transience vs Novelty Plot

```
p_transience <- ggplot(df_res_long, aes(x = surprise, y = transience, color = agent)) +
  geom_point(alpha = 0.5, size = 1.5) +
  geom_smooth(method = "lm", se = TRUE, fullrange = TRUE, size = 1.2) +
  scale_color_manual(values = c("User" = "#377EB8", "LLM" = "#FF7F00")) +
      geom_vline(data = desc_stats,
             aes(xintercept = mean_surprise, color = agent),
             linetype = "dashed", linewidth = 0.8, show.legend = FALSE) +
  labs(
    title = "Transience vs Novelty (Field Data)",
    x = "Novelty (Surprisal)",
    y = "Transience"
  ) +
  theme_minimal(base_size = 14) +
  theme(legend.position = "top")

print(p_transience)
```




```
ggsave(file.path(GRAPHS_DIR, "novelty_transience_vs_novelty.png"),
       p_transience, width = 12, height = 6, dpi = 300)
```

# 3 3. Statistical Models

## 3.1 3.1 Resonance Model

```
model_resonance <- lmer(resonance ~ surprise * agent + (1 | conversation_id), data = df_res_long)
cat("=== Resonance Model ===\n")
```

```
## === Resonance Model ===
```

```
print(summary(model_resonance)$coefficients)
```

```
##                     Estimate Std. Error        df   t value      Pr(>|t|)
## (Intercept)        0.9513552 0.03919177  786.1344 24.274361  1.416786e-97
## surprise           0.9414640 0.02273835 1385.4098 41.404236 1.548277e-244
## agentLLM          -0.6145943 0.06511592 1382.0987 -9.438465  1.545908e-20
## surprise:agentLLM -0.1049326 0.03376284 1383.6349 -3.107931  1.922396e-03
```

```
# Compare slopes
cat("\n=== Slope Comparison (Resonance) ===\n")
```

```
##
## === Slope Comparison (Resonance) ===
```

```
print(emtrends(model_resonance, specs = "agent", var = "surprise"))
```

```
##  agent surprise.trend     SE   df lower.CL upper.CL
##  User           0.941 0.0228 1386    0.897    0.986
##  LLM            0.837 0.0254 1395    0.787    0.886
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
```

### 3.2 3.2 Transience Model

```r
model_transience <- lmer(transience ~ surprise * agent + (1 | conversation_id), data = df_res_long)

cat("=== Transience Model ===\n")
```

```
## === Transience Model ===
```

```r
print(summary(model_transience)$coefficients)
```

```
##                      Estimate Std. Error        df    t value     Pr(>|t|)
## (Intercept)       -0.95135519 0.03919177  786.1344 -24.274361 1.416785e-97
## surprise           0.05853604 0.02273835 1385.4098   2.574331 1.014649e-02
## agentLLM           0.61459428 0.06511592 1382.0987   9.438465 1.545908e-20
## surprise:agentLLM  0.10493258 0.03376284 1383.6349   3.107931 1.922396e-03
```

```r
cat("\n=== Slope Comparison (Transience) ===\n")
```

```
##
## === Slope Comparison (Transience) ===
```

```r
print(emtrends(model_transience, specs = "agent", var = "surprise"))
```

```
##  agent surprise.trend     SE   df lower.CL upper.CL
##  User          0.0585 0.0228 1386   0.0139    0.103
##  LLM           0.1635 0.0254 1395   0.1136    0.213
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
```

# 4 4. Session Information

```r
sessionInfo()
```

```
## R version 4.4.2 (2024-10-31)
## Platform: aarch64-apple-darwin20
## Running under: macOS 26.3
##
## Matrix products: default
## BLAS:   /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRblas.0.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRlapack.dylib;  LAPACK ve
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: Australia/Melbourne
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] emmeans_2.0.1   here_1.0.2      afex_1.4-1      cowplot_1.1.3
##  [5] patchwork_1.3.0 zoo_1.8-12      stringi_1.8.4   arrow_19.0.1.1
##  [9] lmerTest_3.1-3  lme4_1.1-35.5   Matrix_1.7-1    lubridate_1.9.3
## [13] forcats_1.0.0   stringr_1.5.1   dplyr_1.1.4     purrr_1.0.2
```

```
## [17] readr_2.1.5     tidyr_1.3.1     tibble_3.2.1    ggplot2_3.5.1
## [21] tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
##  [1] tidyselect_1.2.1    farver_2.1.2        fastmap_1.2.0
##  [4] TH.data_1.1-2       pacman_0.5.1        digest_0.6.37
##  [7] timechange_0.3.0    estimability_1.5.1  lifecycle_1.0.4
## [10] survival_3.7-0      magrittr_2.0.3      compiler_4.4.2
## [13] rlang_1.1.4         tools_4.4.2         utf8_1.2.4
## [16] yaml_2.3.10         knitr_1.50          labeling_0.4.3
## [19] bit_4.0.5           plyr_1.8.9          abind_1.4-8
## [22] multcomp_1.4-26     withr_3.0.2         numDeriv_2016.8-1.1
## [25] grid_4.4.2          fansi_1.0.6         xtable_1.8-4
## [28] colorspace_2.1-1    scales_1.3.0        MASS_7.3-61
## [31] tinytex_0.52        cli_3.6.3           mvtnorm_1.3-3
## [34] rmarkdown_2.28      crayon_1.5.3        ragg_1.3.2
## [37] generics_0.1.3      rstudioapi_0.16.0   reshape2_1.4.4
## [40] tzdb_0.4.0          minqa_1.2.8         splines_4.4.2
## [43] assertthat_0.2.1    parallel_4.4.2      vctrs_0.6.5
## [46] boot_1.3-31         sandwich_3.1-1      carData_3.0-5
## [49] car_3.1-3           hms_1.1.3           pbkrtest_0.5.3
## [52] bit64_4.0.5         Formula_1.2-5       systemfonts_1.1.0
## [55] glue_1.7.0          nloptr_2.1.1        codetools_0.2-20
## [58] gtable_0.3.5        munsell_0.5.1       pillar_1.9.0
## [61] htmltools_0.5.8.1   R6_2.5.1            textshaping_0.4.0
## [64] rprojroot_2.1.1     vroom_1.6.5         evaluate_1.0.3
## [67] lattice_0.22-6      backports_1.5.0     broom_1.0.6
## [70] Rcpp_1.0.13         coda_0.19-4.1       nlme_3.1-166
## [73] mgcv_1.9-1          xfun_0.54           pkgconfig_2.0.3
```

---

**Analysis complete.** All figures saved to `analysis/figures/` and tables to `analysis/tables/`.